

2020 AMC 10B Problem 20 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 20

Problem:

Let B be a right rectangular prism (box) with edge lengths 1, 3, and 4, together with its interior. For real $r \geq 0$, let $S(r)$ be the set of points in 3-dimensional space that lie within a distance r of some point in B . The volume of $S(r)$ can be expressed as $ar^3 + br^2 + cr + d$, where a , b , c , and d are positive real numbers. What is $\frac{bc}{ad}$?

Answer Choices:

- A. 6
- B. 19
- C. 24
- D. 26
- E. 38

Solution:

Split $S(r)$ into 4 regions:

1. The rectangular prism itself
2. The extensions of the faces of B
3. The quarter cylinders at each edge of B
4. The one-eighth spheres at each corner of B

Region 1: The volume of B is

$$1 \cdot 3 \cdot 4 = 12, \text{ so } d = 12$$

Region 2: This volume is equal to the surface area of B times r (these "extensions" are just more boxes). The volume is then

$$\text{SA} \cdot r = 2(1 \cdot 3 + 1 \cdot 4 + 3 \cdot 4)r = 38r \text{ to get } c = 38$$

Region 3: We see that there are 12 quarter-cylinders, 4 of each type. We have 4 quarter-cylinders of height 1, 4 quarter-cylinders of height 3, 4 quarter-cylinders of height 4. Since 4 quarter-cylinders make a full cylinder, the total volume is

$$4 \cdot \frac{1\pi r^2}{4} + 4 \cdot \frac{3\pi r^2}{4} + 4 \cdot \frac{4\pi r^2}{4} = 8\pi r^2$$

Therefore, $b = 8\pi$.

Region 4: There is an eighth-sphere of radius r at each corner of B . Since there are 8 corners, these eighth-spheres add up to 1 full sphere of radius r . The volume of this sphere is then

$$\frac{4}{3}\pi \cdot r^3, \text{ so } a = \frac{4\pi}{3}$$

Using these values:

$$\frac{bc}{ad} = \frac{(8\pi)(38)}{(4\pi/3)(12)} = (\text{B}) \boxed{19}$$

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